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$f : \mathbb{R} \to \mathbb{R} : x \mapsto x^2$

Stelling : f is continu in $x = 0$

Bewijs:

Pak $\varepsilon > 0$

Stel $\delta = \sqrt{\varepsilon}$

$$\begin{aligned} \forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R} : |x - 0| < \delta &\Rightarrow |f(x) - f(0)| \\ &= |x^2| < \delta^2 = \varepsilon \end{aligned}$$

References